# Course Notes

## Lecture 06 — BLE Gateway, BLE Client and Notifications

*Communication Technologies for the Internet of Things*

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*Text to support independent study*

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## How to Use These Notes

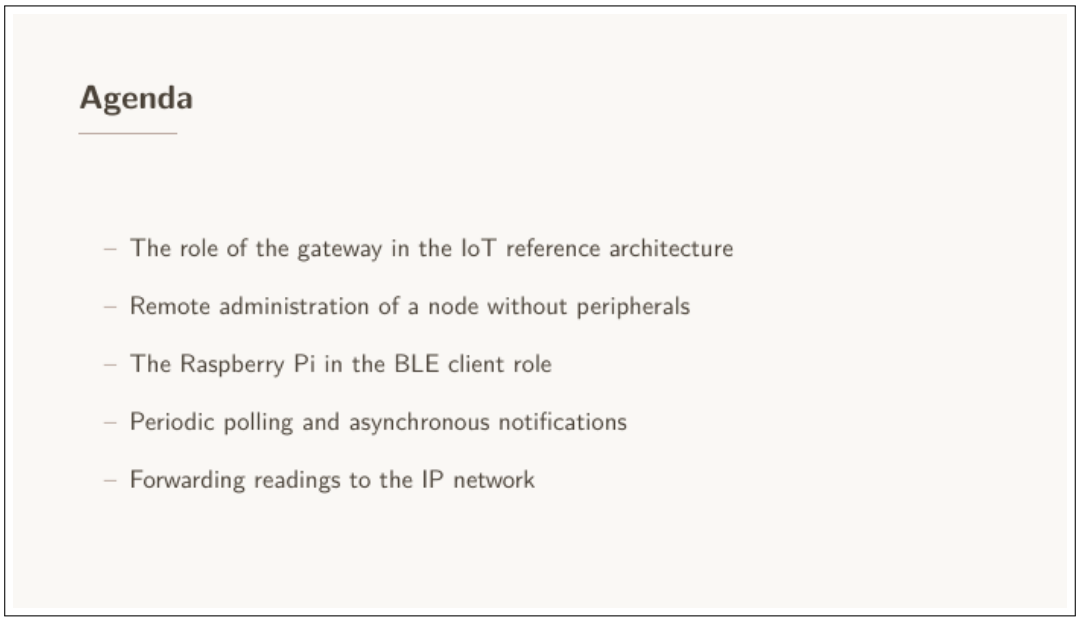
These notes follow the lecture slide by slide. Each page reproduces a slide and states, below it, what should be taken from it.

Reading them does not replace the lecture or the slides. They are meant for later study, once the student has heard the lecture and wants to consolidate, check that the essentials were understood or prepare for assessment. The comments make explicit links between slides that remain implicit in the oral presentation, and point out where the material is commonly misunderstood.

Whenever a comment poses a question, that question is the verification criterion: if you cannot answer it, go back to the slide before moving on.

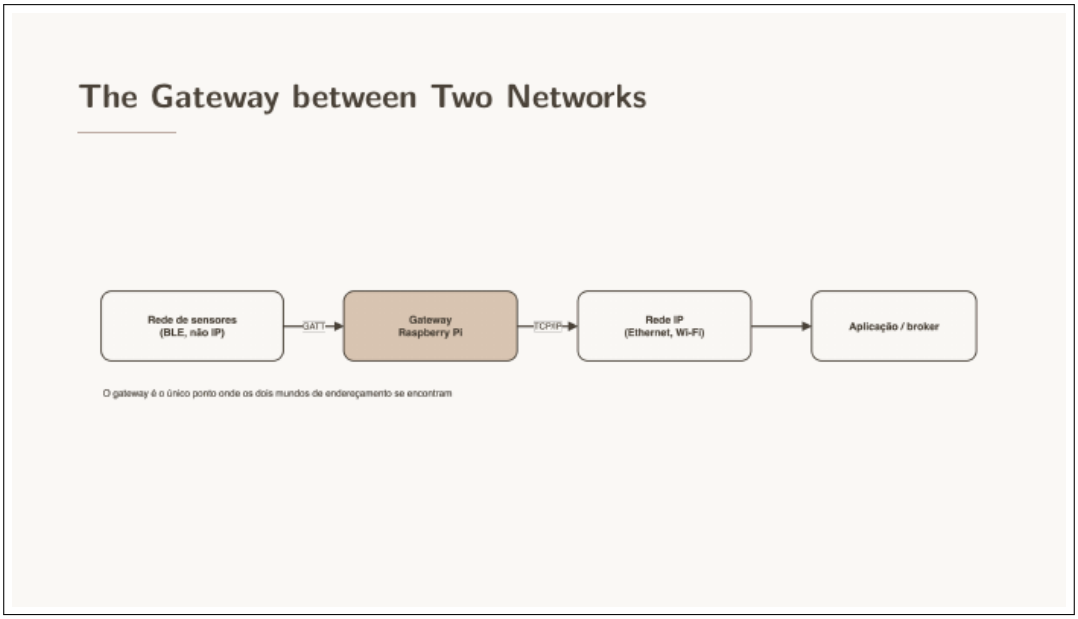
## 1. Commentary on the Slides

### Slide 2. Agenda



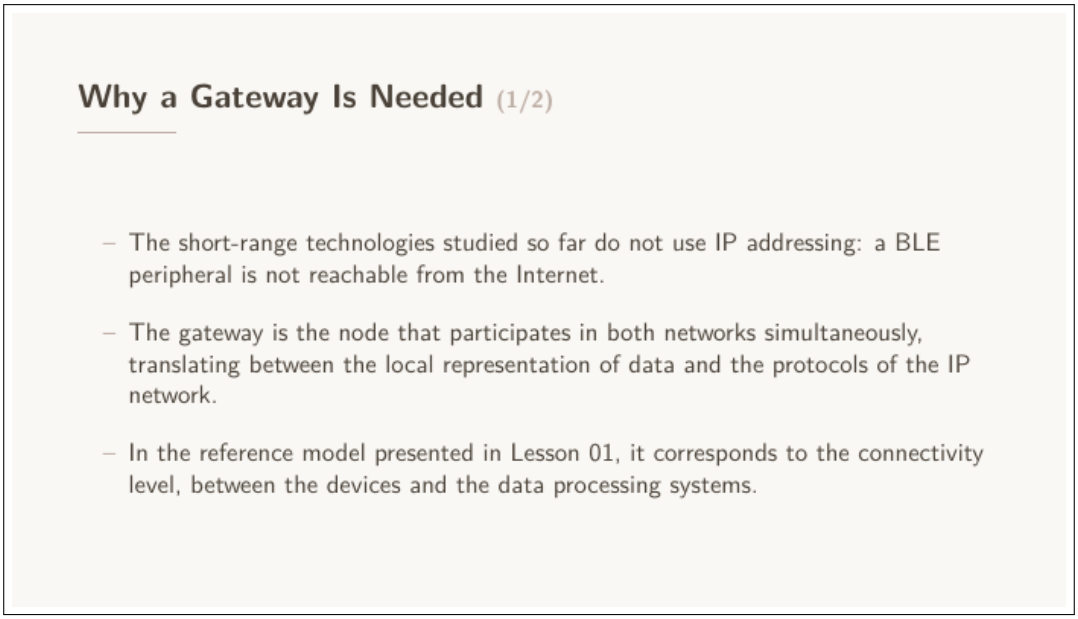

**The path through the lecture.** The lecture answers a single question: how does the data of a sensor that does not speak IP reach an application on the network. The five topics are the five stages of that answer. Note that remote administration appears right after the gateway concept. It is not a side matter: without access to the node, nothing else can be done.

### Slide 3. The Gateway between Two Networks



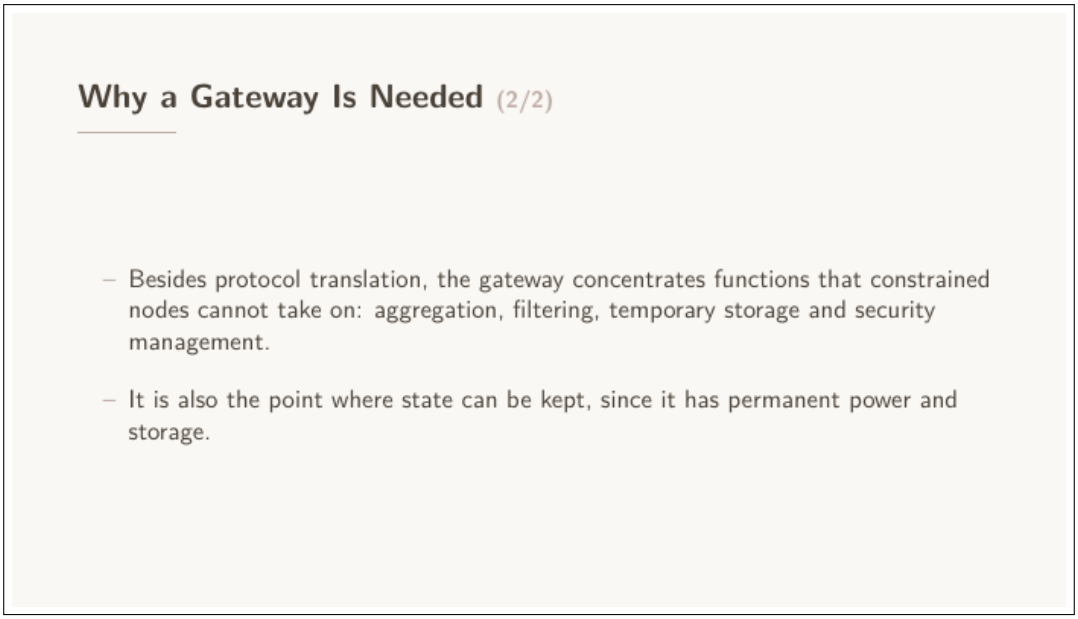

**The two networks and the point of contact.** The diagram is the most important of the lecture. On the left the sensor network, which does not use IP. On the right the IP network and the application. In the middle, a single element belongs to both. The footer sentence deserves attention: the gateway is the only point where the two addressing worlds meet. It follows that it is also a single point of failure, a question taken up in the lab session.

## Slide 4. Why a Gateway Is Needed (1/2)



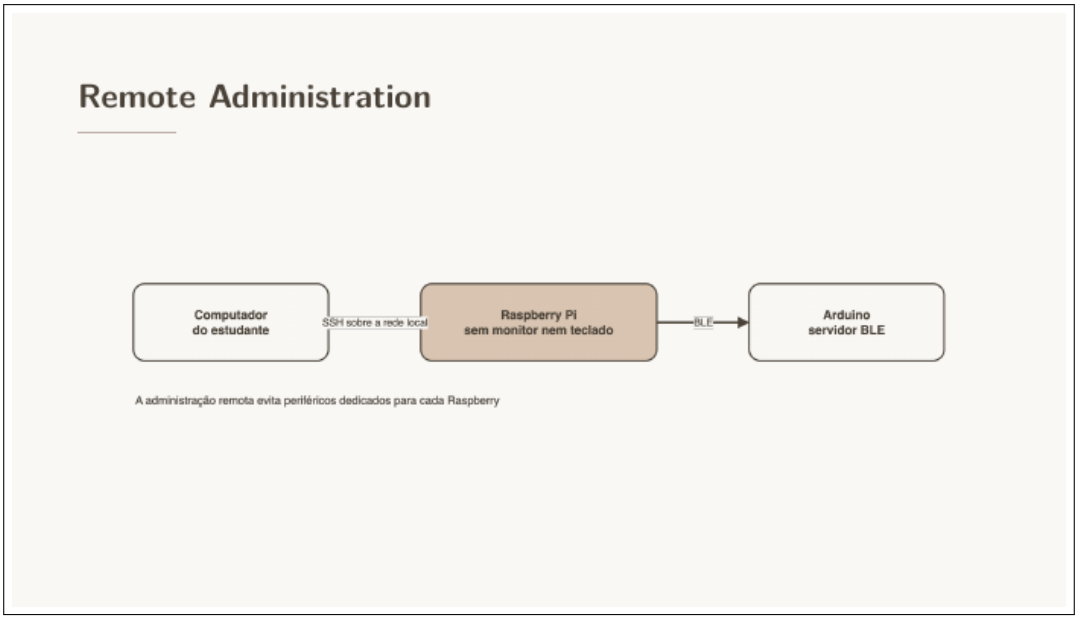

**Because the peripheral is not reachable.** A BLE peripheral has no IP address and is not reachable from the Internet. This is the technical reason for the gateway's existence. In the reference model presented in Lecture 01, the gateway corresponds to the connectivity level, between devices and data processing systems. It is worth going back to that model and locating the element there.

## Slide 5. Why a Gateway Is Needed (2/2)



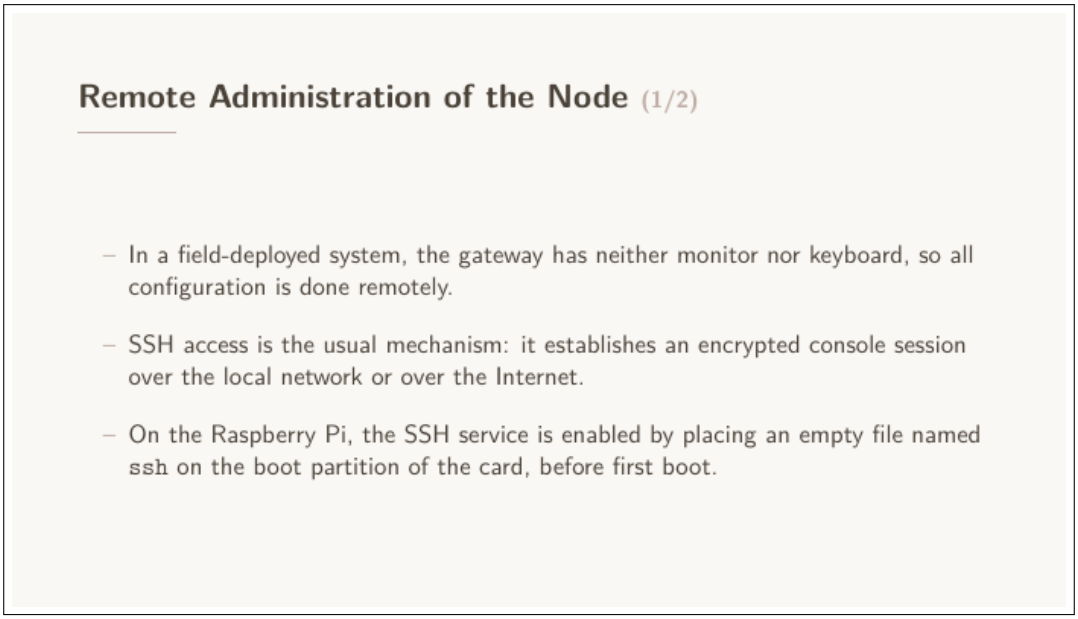

**What else the gateway does.** Besides protocol translation, the gateway concentrates functions that constrained nodes cannot take on: aggregation, filtering, temporary storage and security management. It is also the point where state can be kept, since it has permanent power and storage. A battery-powered sensor cannot keep history; the gateway can.

### Slide 6. Remote Administration



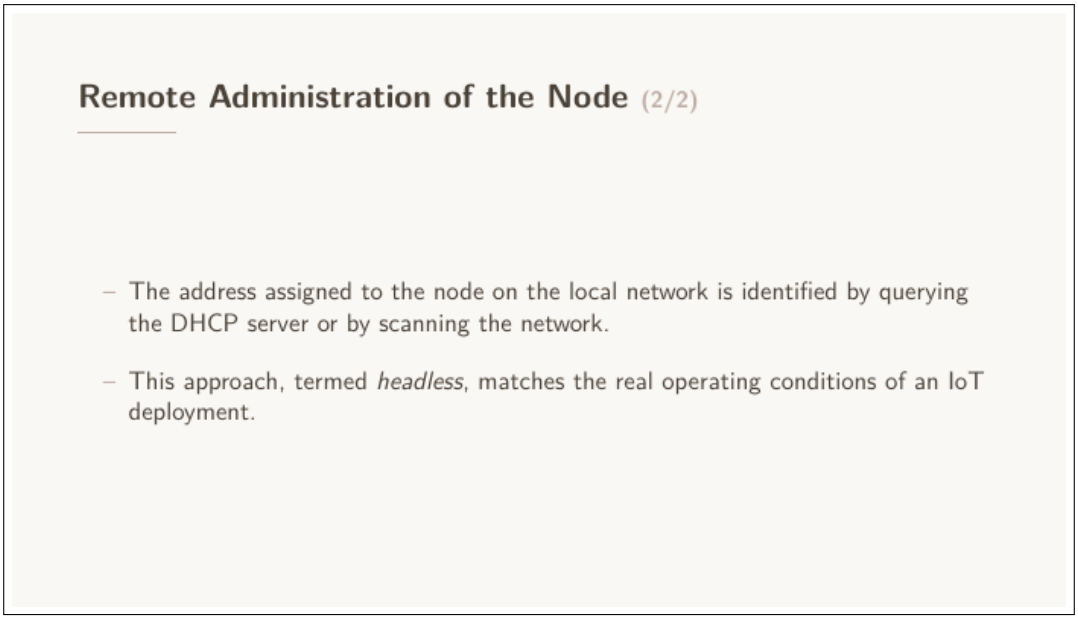

**The real configuration of an installed node.** The diagram shows the working topology: the student's computer accesses the Raspberry Pi by SSH, and the Pi communicates by BLE with the Arduino. The footnote explains the choice: remote administration avoids dedicated peripherals for each node, which in an installation with dozens of gateways is decisive.

### Slide 7. Remote Administration of the Node (1/2)



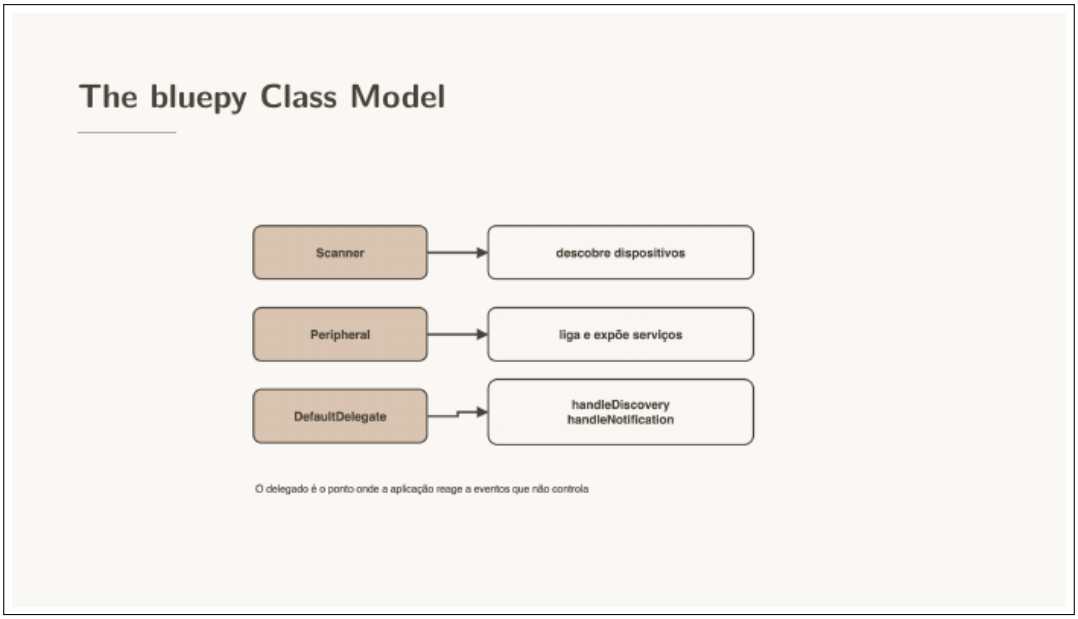

**Access with no monitor or keyboard.** In a system installed in the field, the gateway has no peripherals. All configuration is done remotely, and SSH is the usual mechanism, since it establishes an encrypted console session. On the Raspberry Pi, the service is enabled by placing an empty file named ssh on the card's boot partition, before the first boot.

### Slide 8. Remote Administration of the Node (2/2)



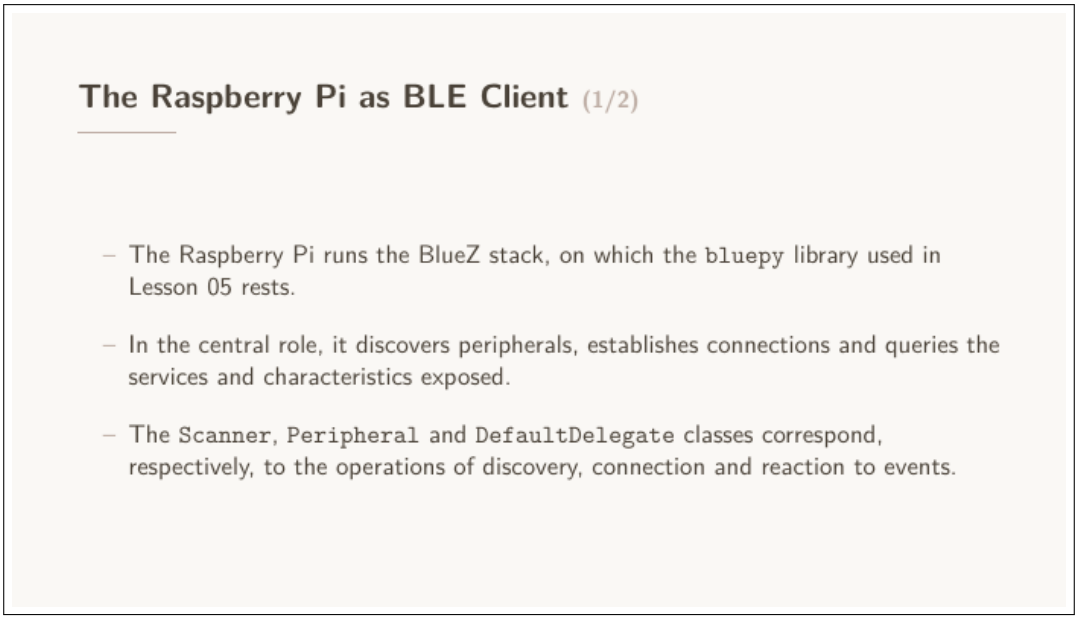

**Finding the node on the network.** The address assigned is identified by consulting the DHCP server or by scanning the network. This approach, called headless, matches the real operating conditions of an IoT installation, which is why it is adopted from the first session.

### Slide 9. The bluepy Class Model



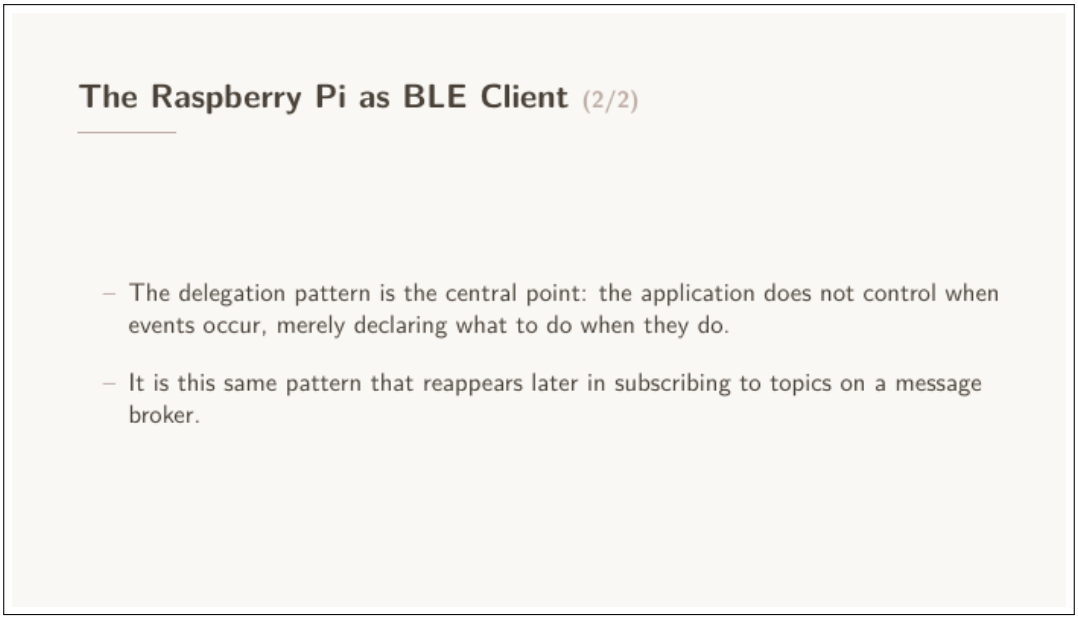

**The three classes that matter.** The diagram recalls the model from Lecture 05. Scanner discovers, Peripheral connects and exposes services, DefaultDelegate reacts to events. The footer sentence identifies the conceptual point: the delegate is where the application reacts to events it does not control.

### Slide 10. The Raspberry Pi as BLE Client (1/2)



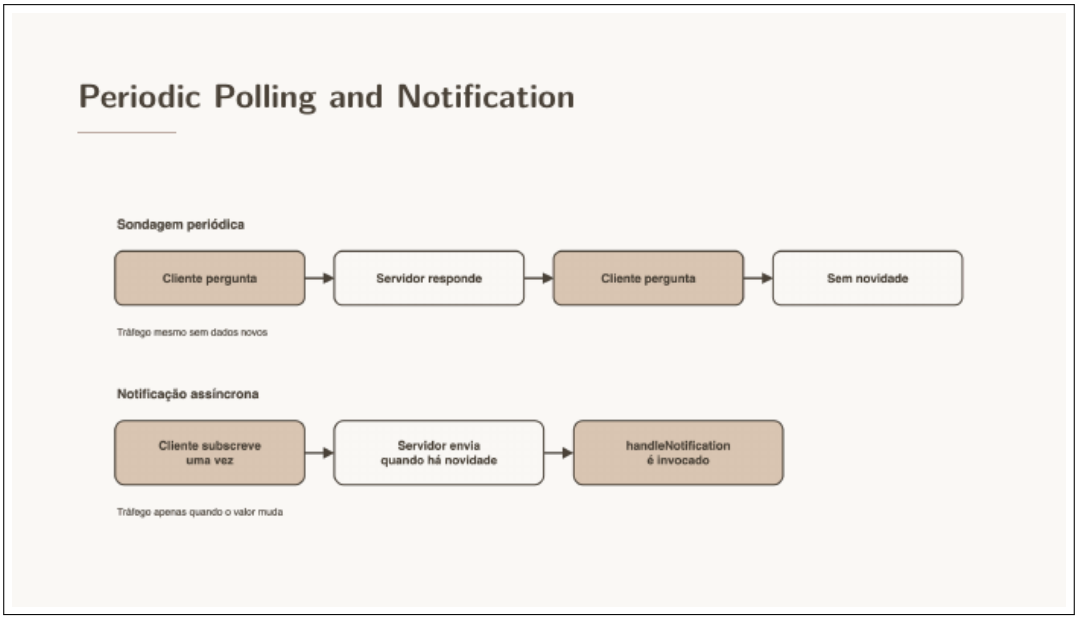

**The Raspberry Pi in the central role.** The Raspberry Pi runs the BlueZ stack, on which bluepy sits. In the central role, it discovers peripherals, establishes the connection and queries the services exposed. The three classes correspond, respectively, to discovery, connection and reaction to events.

### Slide 11. The Raspberry Pi as BLE Client (2/2)



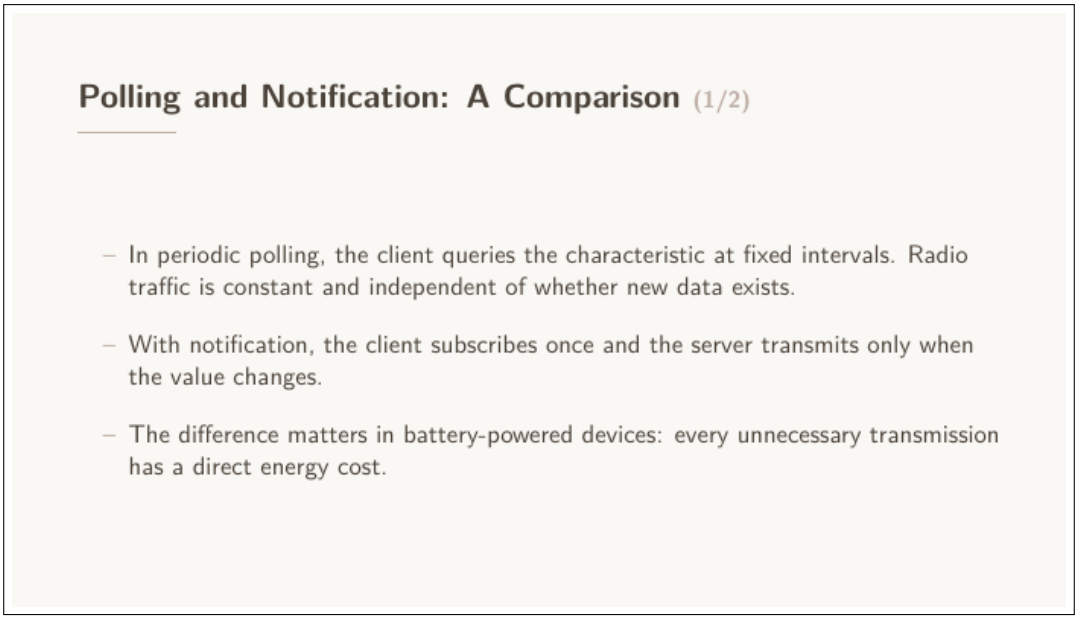

**A pattern that repeats.** The central point is the delegation pattern: the application does not control the instant at which events occur, and just declares what to do when they do. It is worth fixing this formulation, because the same pattern reappears in topic subscription on a message broker, in Lecture 12. Recognising the repetition saves study effort.

## Slide 12. Periodic Polling and Notification



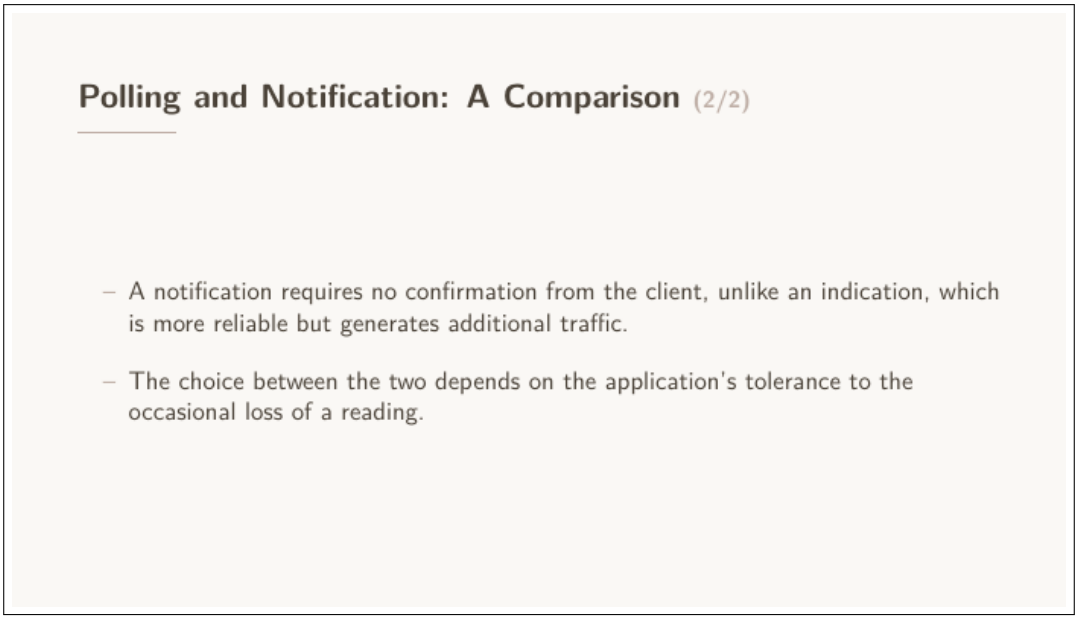

**The two regimes, side by side.** In the upper half, periodic polling: the client asks, the server answers, and sometimes there is no news at all. In the lower half, notification: the client subscribes once and the server transmits when the value changes. Note the footer captions of each half. They summarise the difference better than any description.

### Slide 13. Polling and Notification: A Comparison (1/2)



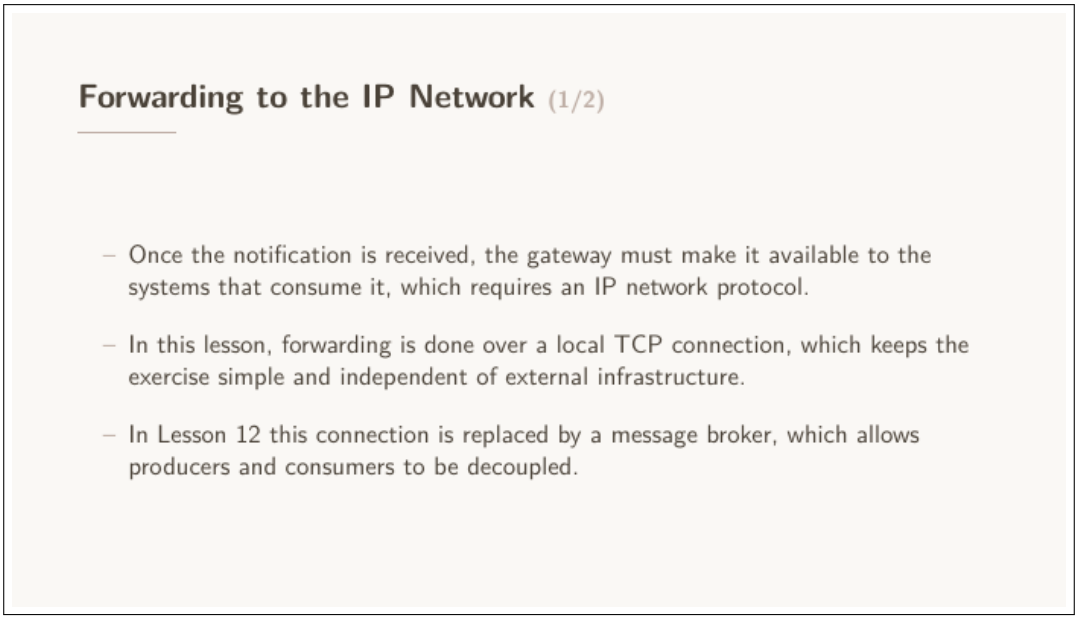

**Why the difference matters.** With polling, radio traffic is constant and independent of whether there is new data. With notification, there is transmission only when the value changes. The consequence is direct on battery-powered devices: each unnecessary transmission has an energy cost. For a sensor that reports one change per hour, the difference between the two regimes is several orders of magnitude.

### Slide 14. Polling and Notification: A Comparison (2/2)



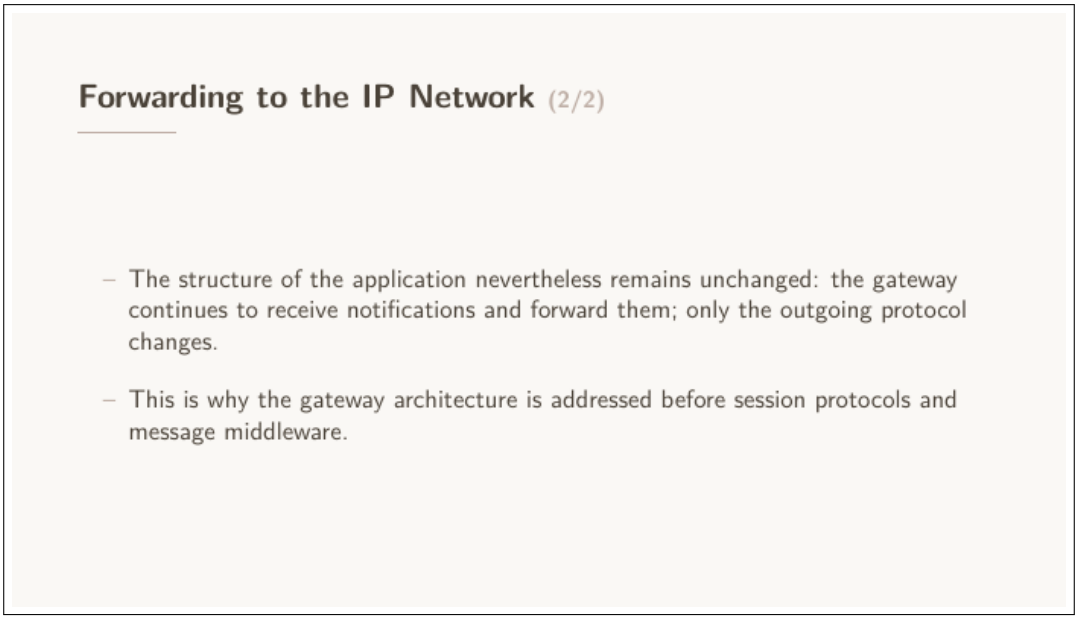

**When to choose each.** Notification requires no acknowledgement from the client, unlike indication, which is more reliable but generates additional traffic. The choice depends on the application's tolerance of the occasional loss of a reading. A temperature sensor tolerates it; an intrusion alarm does not.

### Slide 15. Forwarding to the IP Network (1/2)



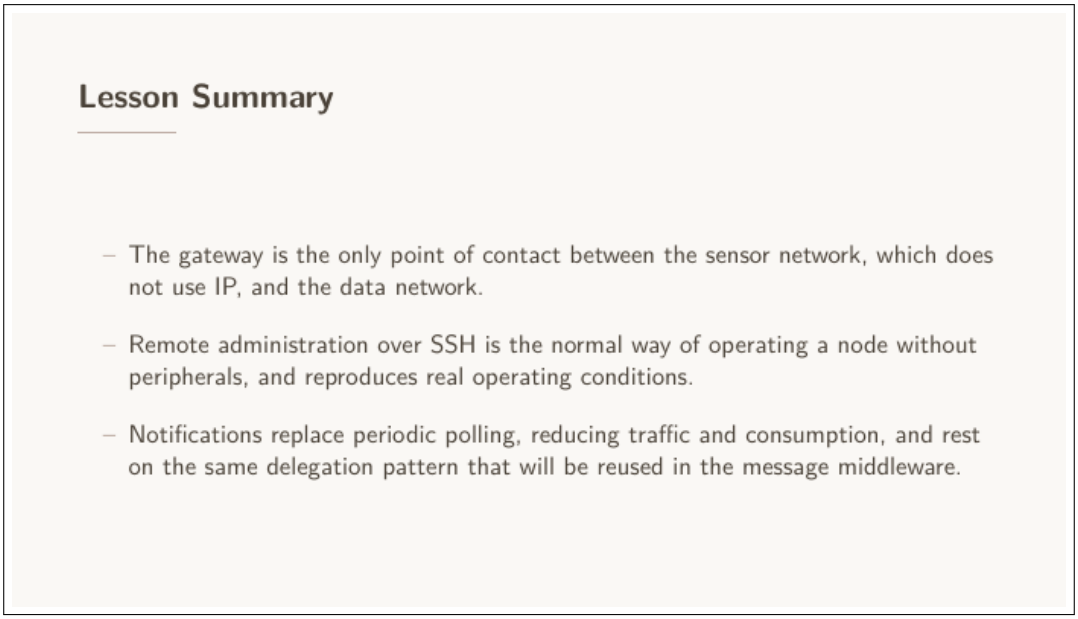

**From BLE to the network.** Once the notification is received, the gateway has to make it available to the systems that consume it, which requires an IP network protocol. In this lecture a local TCP connection is used, which keeps the exercise simple and independent of external infrastructure. In Lecture 12 that connection is replaced by a message broker.

### Slide 16. Forwarding to the IP Network (2/2)



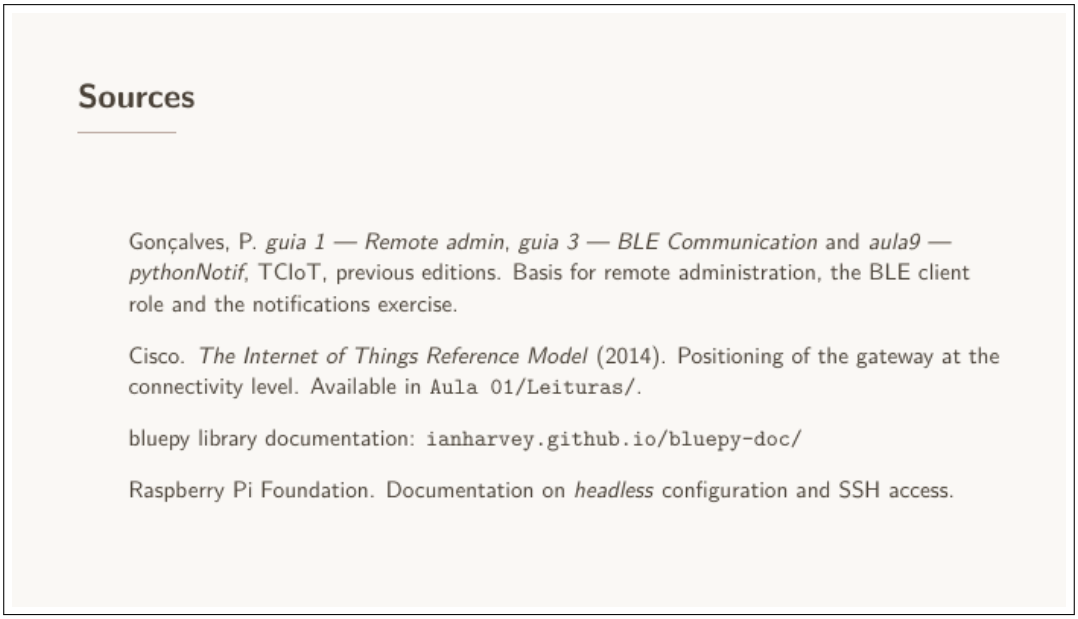

**What changes and what stays.** The structure of the application remains unchanged: the gateway keeps receiving notifications and forwarding them. Only the outgoing protocol changes. This is why the gateway architecture is covered before session protocols and message middleware. Once the structure is understood, replacing the protocol is a detail.

### Slide 17. Lesson Summary




**Lecture summary.** The gateway is the only point of contact between the sensor network and the data network. Remote administration by SSH reproduces real operating conditions. Notifications replace periodic polling and rest on the same delegation pattern that will be reused in message middleware.

## Slide 18. Sources




**Origin of the material.** The basis is the lab guides of the course from previous editions, which had no associated theory component. The framing of the gateway draws on the reference model published by Cisco, already used in Lecture 01 and available in its readings folder.

## Final Check

Before considering the material studied, check that you can answer the following without looking at the slides.

1. Explain why a BLE peripheral is not reachable from the Internet.
2. Locate the gateway in the reference model presented in Lecture 01.
3. List functions the gateway takes on that a constrained node cannot, with justification.
4. Describe the procedure to enable SSH access on a Raspberry Pi with no peripherals.
5. Compare the traffic generated by periodic polling and by notification on a sensor that reports one change per hour.
6. State in which circumstance indication should be preferred to notification.
7. Explain what changes, and what stays, when the local TCP connection is replaced by a message broker.

## References


- Gonçalves, P. *guide 1 — Remote admin*, *guide 3 — BLE Communication* and *aula9 — pythonNotif*, TCIoT, previous editions.
- Cisco. *The Internet of Things Reference Model* (2014). Available in `Aula 01/Leituras/`.
- bluepy library documentation: `ianharvey.github.io/bluepy-doc/`
- Raspberry Pi Foundation. Documentation on *headless* configuration and SSH access.